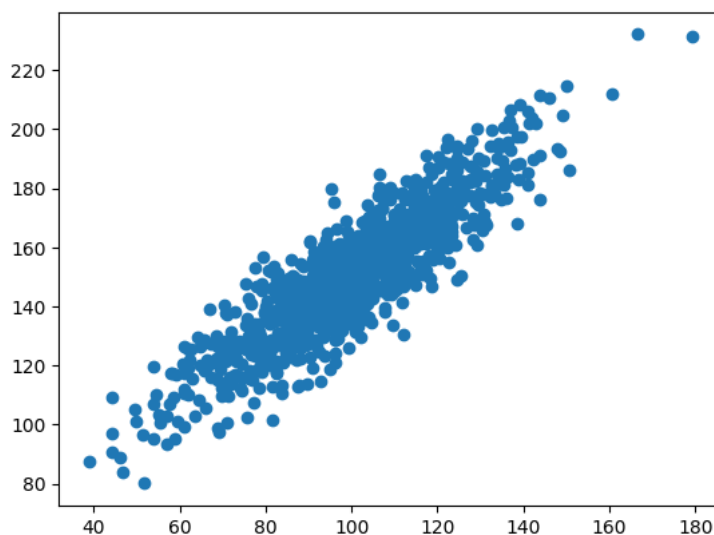


1. Covariance Pearson Correlation

The first notebook presents two related variables.

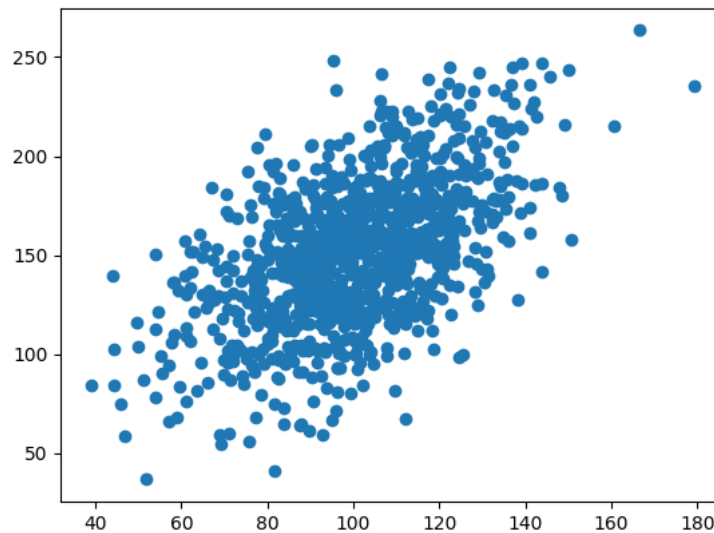
- Correlation is 0.888, showing a strong positive relationship
- Covariance is 389, which is still positive
- The scatter plot shows numerous points forming an upward trend (positive correlation)



Increasing the noise

- Changed the line $\text{data2} = \text{data1} + (10 * \text{randn}(1000) + 50)$ to $\text{data2} = \text{data1} + (30 * \text{randn}(1000) + 50)$
- The Pearson correlation is 0.56, meaning that the relationship is weaker
- Covariance is 398 is still positive, but less informative about the strength

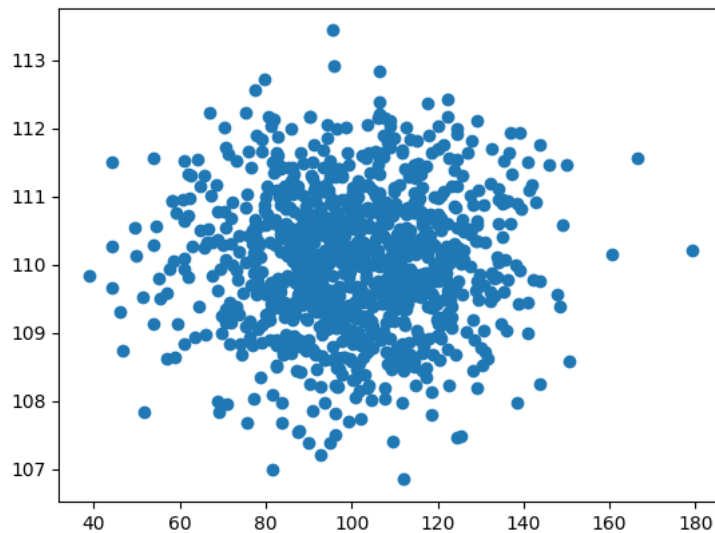
- Scatter plot: the points are more dispersed, although a general upward trend is still visible



Removing the dependency between variables

- Changed the line $\text{data2} = \text{data1} + (10 * \text{randn}(1000) + 50)$ to $\text{data2} = 20 + \text{randn}(1000) + 100$
- Pearson Correlation is 0.02, indicating no linear relationship between variables.
- The covariance is 0.44, which is close to zero, confirming a weak association

Scatter plot: shows a completely random distribution of points.



Observations

This activity shows that correlation and covariance depend strongly on data structure. When noise increases, correlation decreases. When the variables are independent, correlation becomes meaningless. This highlights the importance of data quality in ML modeling.

2. Linear regression code

The second notebook presents a linear regression.

The first part of the code is used to define whether there is a correlation between x and y . The Pearson correlation is -0.759 , meaning that the correlation is strong and negative (when x increases, y tends to decrease significantly).

The second part of the code predicts future values using a mathematical model and is used to estimate the y values based on new x values ($x = 10$).

Transform data into perfect data

The arrays representing the values of the x - and the y - axes change from

$x = [5, 7, 8, 7, 2, 17, 2, 9, 4, 11, 12, 9, 6]$

$y = [99, 86, 87, 88, 111, 86, 103, 87, 94, 78, 77, 85, 86]$

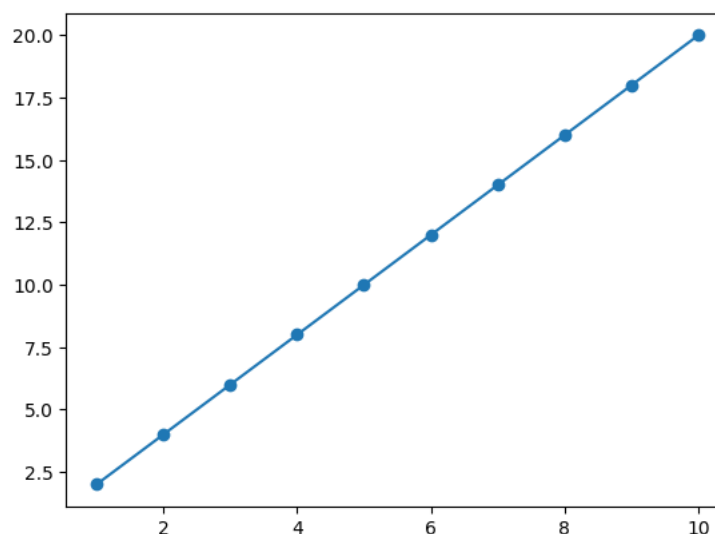
to

$x = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]$

$y = [2, 4, 6, 8, 10, 12, 14, 16, 18, 20]$

The Pearson correlation changed to 1.000 , meaning that the correlation is positive and perfect. When X increases, Y increases proportionally.

From the graph, it is possible to observe that all the points lie on a straight line.



Prediction change

In the second part of the script, I changed the command “speed = myfunc(10)” to “speed = myfunc(30)” while the x and y variables are from the original example.

The result changes from 85.59 to 50.57 because changing the input from 10 to 30 moves along the same regression line, resulting in different y values. Only the prediction point changes.

Observation

In conclusion, linear regression helps to understand and predict relationships between variables. When data became perfectly linear, the correlation became 1.000. Changing the input value only changes the predicted result, not the model.

3. Multiple Linear Regression

The third code presents a Multiple Linear Regression, which uses multiple independent variables together to predict an outcome.

In this specific case, the two variables used to predict CO₂ emissions are weight and volume. Since the original “car.csv” dataset was not available, the following observations are based on the expected behaviour of a multiple regression model.

Change the variables

The original weight and volume used for the prediction were 2300 kg and 1300 cm³, respectively. If the weight is increased from 2300 to 3300 kg while the volume is kept constant, it is expected that the predicted CO₂ emissions will increase because heavier vehicles usually require more energy to move.

If the weight remains at its original value (2300 kg) but the volume is increased by a similar amount, the predicted CO₂ emissions are expected to increase because larger engines typically consume more fuel.

If both the volume and weight are reduced, the predicted CO₂ emissions will decrease.

Observation:

The predicted CO₂ emissions vary when the input variables (weight and volume) are changed. In general, increasing either weight or volume leads to higher predicted CO₂ emissions, whereas decreasing these variables leads to lower predicted CO₂ emissions.

4. Polynomial Regression

The last exercise presents a Polynomial Regression model. The model predicts the speed of a vehicle at different time points, where x represents time and y represents speed.

The model creates a polynomial curve for the data, allowing it to capture non-linear relationships between the variables. It also provides an R^2 value that shows how well the model fits the data, which is 0.94 in this case, indicating that the model fits the data well.

Finally, the model predicts values for inputs that are not included in the dataset, such as $x = 17$.

Changing y values

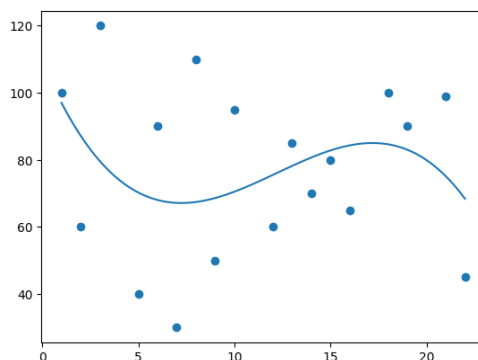
To start, the y -values were modified to increase the noise.

From

$y = [100, 90, 80, 60, 60, 55, 60, 65, 70, 70, 75, 76, 78, 79, 90, 99, 99, 100]$

to

$y = [100, 60, 120, 40, 90, 30, 110, 50, 95, 60, 85, 70, 80, 65, 100, 90, 99, 45]$



The output is a less precise curve with more scattered points and an extremely low R^2 (0.11), indicating that there is no strong relationship.

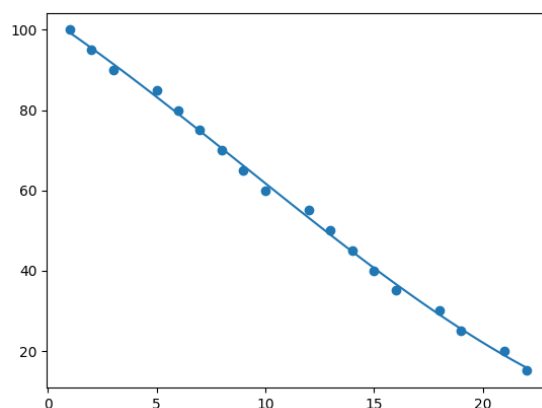
Then, the y-values were replaced.

From

$y = [100, 60, 120, 40, 90, 30, 110, 50, 95, 60, 85, 70, 80, 65, 100, 90, 99, 45]$

to

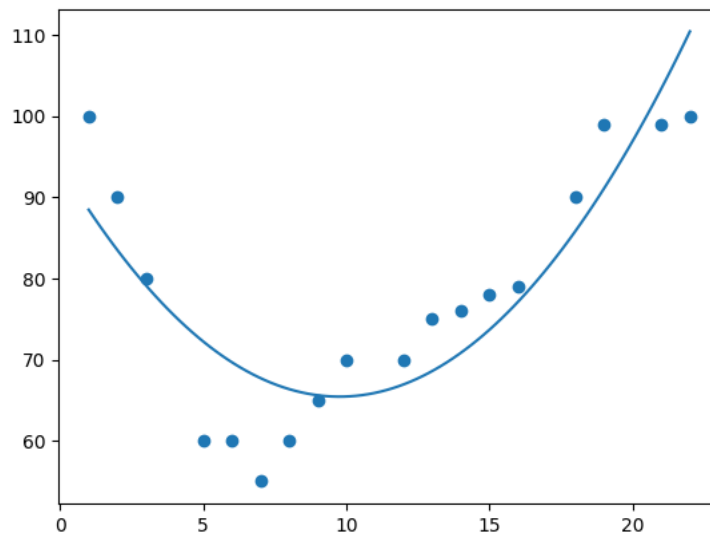
$y = [100, 95, 90, 85, 80, 75, 70, 65, 60, 55, 50, 45, 40, 35, 30, 25, 20, 15]$



The output is a straight line with the points lying very close to it, indicating a strong relation confirmed by the R^2 value of 0.99.

Changing the polynomial

Then I decided to change the polynomial grade from 3 to 2. As a result, I obtained a graph with a simpler curve and less flexibility, and an R^2 of 0.76, which means that the model is still good but less so with a polynomial grade of 3.



Change the prediction point

Finally, I changed the prediction point to observe the changes.

The original prediction point was 17, and the predicted speed was 88.87. When the prediction point was changed to 2, the predicted speed became 87.82, whereas when the prediction point was changed to 22, the predicted speed became 99.24.

Observation

Changing the y-values affects how well the polynomial model fits the data. A more regular dataset resulted in a more accurate curve, while noisy data reduced the model's performance.

Modifying the polynomial degree from 3 to 2 reduced the model's performance. When the polynomial degree is 2, it produces a smoother curve but may not capture all variations in the data as accurately as the cubic model. As a result, the model may achieve slightly lower accuracy.

Finally, changing the prediction point results in a different output because the model estimates values based on the shape of the fitted curve. Each input value corresponds to a different position on the curve, therefore producing different predicted values.